

Mechanical Design Evolution: Hull Selection Analysis (SWATH vs. Catamaran)

RC Boat Water Quality Project

Abstract

This report documents the mechanical design evolution of the autonomous water quality monitoring RC boat. Initially conceptualized as a Small Waterplane Area Twin Hull (SWATH) vessel, the design underwent a rigorous review following consultation with maritime industry professionals. Based on the specific operational constraints of navigating the River Beas and carrying a heavy sensor payload, the design was officially pivoted to a Catamaran hull. This report outlines the comparative analysis that drove this engineering decision.

1 Hull Design Concepts

- **Catamaran:** Features two parallel hulls that sit directly on the surface of the water. The hulls are typically narrow and provide buoyancy evenly across the entire length of the waterline.
- **SWATH (Small Waterplane Area Twin Hull):** Also utilizes two hulls, but the primary buoyancy is provided by fully submerged, torpedo-like pontoons. These pontoons are connected to the main payload platform above the water by very thin vertical struts, intentionally keeping the “waterplane area” (the part intersecting the surface) as small as possible.

2 Comparative Analysis

The two hull designs were evaluated across four key hydrodynamic and operational metrics:

2.1 1. Seakeeping and Stability

- **Catamaran:** Because the hulls sit on the surface, a catamaran tends to follow the contour of the waves. In rough seas, the large waterplane area reacts heavily to wave energy, leading to pitching and rolling.
- **SWATH:** SWATH excels in rough water. Because the primary buoyancy is well below surface wave action, and the waterline struts are incredibly thin, waves pass right through the structure without lifting it, resulting in a remarkably stable platform.

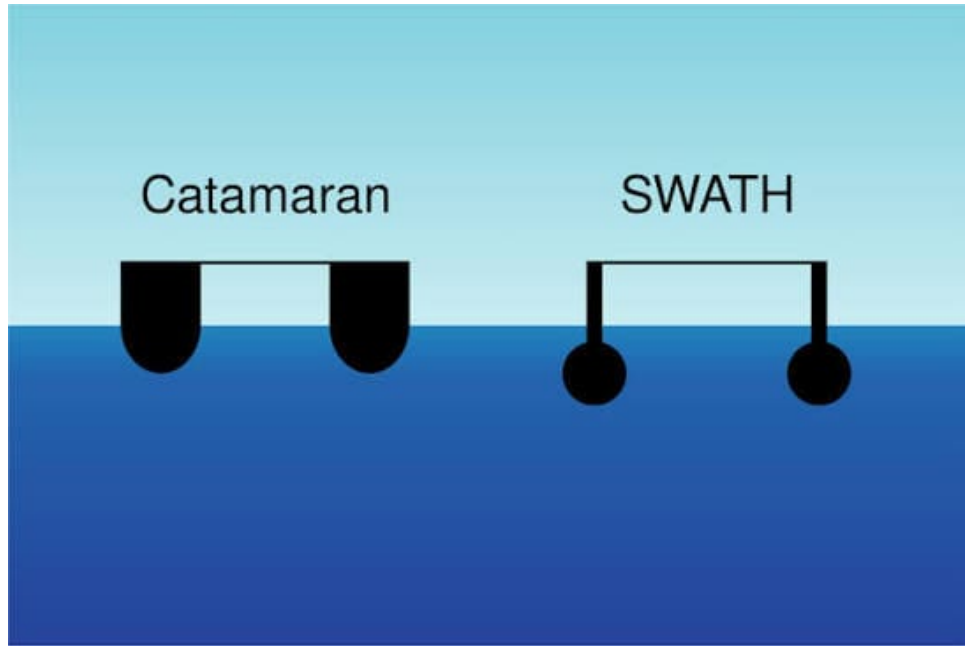


Figure 1: Visual Comparison of Catamaran and SWATH Hull Cross-sections.

2.2 2. Speed and Efficiency

- **Catamaran:** In calm conditions, catamarans are highly efficient. Their slender hulls have a low “wetted surface area,” which significantly reduces drag.
- **SWATH:** SWATHs are generally slower and less energy-efficient. The deeply submerged pontoons and struts create a large wetted surface area, resulting in high frictional drag that requires significantly more horsepower to overcome.

2.3 3. Draft (Depth in Water)

- **Catamaran:** Known for having a very shallow draft, allowing operations in shallow coastal waters, rivers, and close to riverbanks.
- **SWATH:** Has a deep draft because the large buoyancy pontoons must be submerged entirely below the wave action, restricting operation in shallow areas.

2.4 4. Sensitivity to Weight and Load

- **Catamaran:** Can carry significant loads. While added weight pushes the boat deeper, the large waterplane area easily handles the added displacement without drastically changing the boat’s profile.
- **SWATH:** Highly sensitive to weight changes. Because only the thin struts intersect the waterline, adding even a small amount of weight causes the boat to sink significantly deeper. Maintaining proper ride height in a SWATH often requires complex, active ballast pumping systems.

3 Rationale for Shifting to Catamaran

While the SWATH design initially seemed appealing for its superior stability, consultation with maritime professionals revealed that it was fundamentally mismatched for our specific use-case on the River Beas. The pivot to a Catamaran was executed based on the following engineering realities:

1. **Shallow Water Navigation (Draft):** The River Beas features shallow, rocky banks and variable depths. A SWATH's deep draft would lead to frequent beaching and potential catastrophic damage to the submerged sensor payload. The Catamaran's shallow draft ensures safe operation across all river zones.
2. **Payload Sensitivity (Weight):** Our RC boat carries a dense, heavy payload including lithium batteries, an STM32 microcontroller enclosure, solar panels, and 6 heavily calibrated sensors (pH, DO, Turbidity, EC, ORP, Temp). A SWATH hull would sink too deeply under this load and require a complex active ballast system. The Catamaran effortlessly displaces this weight while maintaining a safe ride height.
3. **Energy Efficiency (Drag):** The boat must operate autonomously on limited battery power. The calm-water efficiency and low frictional drag of the Catamaran extend the operational range significantly compared to the high-drag SWATH design.
4. **Manufacturing Feasibility:** 3D printing, assembling, and properly waterproofing the complex submerged pontoons and thin struts of a SWATH at a small RC scale is highly prone to failure. The Catamaran offers a massive, flat deck space that simplifies mounting the electronics box and routing sensor cables down to the water.

4 Conclusion

By shifting from the SWATH to the Catamaran design, the project sacrificed extreme rough-water stability—which is largely unnecessary for standard river monitoring—in exchange for crucial gains in payload capacity, shallow-water capability, energy efficiency, and manufacturing feasibility.